



witness

F class defines geometry

IPMs: Definition

- An Integral Probability Metric:

$$d_{\mathcal{F}}(\alpha, \beta) = \sup_{f \in \mathcal{F}} \left(\mathbb{E}_{\alpha} f - \mathbb{E}_{\beta} f \right) = \sup_{f \in \mathcal{F}} \int_{\mathcal{X}} f(x) \, d\alpha(x) - \int_{\mathcal{X}} f(x) \, d\beta(x)$$

where $\mathcal{F}$ is some function class

- $f : \mathcal{X} \rightarrow \mathbb{R}$ is called the ‘**witness**’ function
- Choice of $\mathcal{F}$ **class defines geometry** / form of the IPM

IPMs: Intuition

$$d_{\mathcal{F}}(\alpha, \beta) = \sup_{f \in \mathcal{F}} \int_{\mathcal{X}} f(x) \, d\alpha(x) - \int_{\mathcal{X}} f(x) \, d\beta(x)$$